

Worksheet · Section 1.4

Identifying & Correcting Errors



The Big Picture

Every program has bugs at first. Practice naming the four error types, tracing code by hand, and choosing the test cases that catch bugs — the exam loves “which input reveals the error?”

Part 1 · Vocab Match

Write the letter of the matching definition on each line.

- | | |
|----------------------------------|---|
| _____ 1. Syntax error | A. Tracking variable values step by step on paper |
| _____ 2. Logic error | B. The rules of the programming language are not followed |
| _____ 3. Run-time error | C. An input at or just beyond the edge of the valid range |
| _____ 4. Overflow error | D. An error that happens while the program is running |
| _____ 5. Debugging | E. A tool that pauses a program and steps through it line by line |
| _____ 6. Test case | F. A specific input with an expected output used to check a program |
| _____ 7. Boundary case | G. A number is outside the range a fixed number of bits can hold |
| _____ 8. Hand tracing | H. A mistake in the algorithm that causes incorrect or unexpected behavior |
| _____ 9. Debugger | I. What a correct program should produce for a given input |
| _____ 10. Expected output | J. Finding and fixing errors in a program |

Part 2 · Fill It In

Use each word from the word bank once.

syntax**overflow****trace****run-time****logic****boundary****debugger****test case**

1. Forgetting a closing parenthesis causes a _____ error.
2. A program that averages scores but divides by the wrong number has a _____ error.
3. Trying to access index 0 of an AP list causes a _____ error.
4. Storing 300 in an 8-bit number causes an _____ error.
5. For a rule that accepts ages 13 to 19, the input 13 is a _____ value.
6. Running a program with input 5 and checking that it outputs 25 is a _____.
7. Writing each variable's value after every line is how you _____ code.
8. A _____ lets you pause a program and run it one line at a time.

Name: _____

Class: _____

Date: _____

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Practice

Part 3 · Name That Error

Check the type of error for each situation.

Item	Syntax	Run-time	Logic	Overflow
1. DISPLAY("Hi"				
2. $x \leftarrow 10 / 0$				
3. A tip calculator multiplies by 1.5 instead of 0.15				
4. A score stored in 8 bits goes from 255 to 0				
5. IF ($x > 5$ {				
6. Accessing list[LENGTH(list) + 1]				
7. A find-the-max procedure returns 0 for [-3, -1, -7]				
8. A 4-bit counter adds 1 to 15				

Part 4 · Trace It

```

total ← 0
n ← 4
REPEAT UNTIL (n = 0)
{
    total ← total + n
    n ← n - 1
}
DISPLAY(total)

```

Pass	total	n
start	0	4
1	_____	_____
2	_____	_____
3	_____	_____
4	_____	_____

Displayed: _____. If n started at -1, the loop would be _____ because n never equals 0.

Name: _____

Class: _____

Date: _____

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Apply It

Part 5 · Pick the Test Cases

An online store gives free shipping when the order total is from \$50 to \$500, inclusive.

Four boundary tests: _____, _____, _____, _____.

If the code uses `total > 50`, which input reveals the bug? _____

Name one invalid input to test:

Part 6 · Debug the Procedure

This should count scores of **70 or higher**. Find TWO errors and fix them.

```
PROCEDURE countPassing(scores)
{
  count ← 0
  FOR EACH s IN scores
  {
    IF (s > 70)
    { count ← count + 1 }
    RETURN (count)
  }
}
```



Think:

Test the edges! Most bugs hide at the boundaries: the first value, the last value, and the values just outside the valid range.

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Exam Practice

Part 7 · Exam Practice

★ Warm-up

1. Which type of error is usually found **before** a program runs?

- A. Syntax error
- B. Run-time error
- C. Overflow error
- D. Logic error

★★ On Level

2. A program displays "Teen" for age 19 but not for age 13. What is most likely?

- A. A run-time error
- B. A syntax error
- C. A logic error in the condition
- D. An overflow error

★★ On Level

3. What happens when this code runs on a non-empty list?

```
i ← 1
REPEAT UNTIL (i > LENGTH(lst))
{ lst[i] ← lst[i] * 2 }
```

- A. An infinite loop occurs because i never changes
- B. Every element is doubled
- C. Only the first element is doubled
- D. A syntax error occurs

★★★ Exam Hard

4. Which test list is BEST for checking a procedure that finds the minimum value?

- A. [4]
- B. [5, -2, 0, -9]
- C. [1, 2, 3]
- D. [10, 20, 30]

★★★ Exam Hard

5. Which TWO strategies help find a logic error? (*Select two.*)

- A. Hand tracing with a trace table
- B. Renaming variables to single letters
- C. Adding DISPLAY statements to show variable values
- D. Deleting all comments

Part 8 · Written Response

A classmate's grade-average program works on their test data, but a teacher reports wrong averages. Describe a plan to find and fix the error using at least two debugging strategies.
